

NETWORKS AND COMPLEXITY

Exercise Sheet 14: Complexity-Stability Debate

This is an exercise sheet from the forthcoming book Networks and Complexity.

Find more exercises and solutions at <https://github.com/NC-Book/NCB>

Ex 14.1: Push and Pull [Lvl. 1]

Consider a steady state with the abstract Jacobian

$$\mathbf{J} = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix}$$

Compute the eigenvalues of this matrix to show: a) The off-diagonal elements only in the form of the weight of the cycle $w = J_{12}J_{21}$ matters for the eigenvalues, b) if $w = 0$ the eigenvalues are J_{11} and J_{22} . c) if $w < 0$ the eigenvalues are pulled together, d) if $w > 0$ the eigenvalues are pushed apart.

Ex 14.2: Stability analysis in one dimension [Lvl. 2]

Find the steady states of the following one-dimensional systems and determine their stability:

a)

$$\dot{x} = 5 - 3x$$

b)

$$\dot{x} = (3 - x)(x + 5)(x - 9)(x - 1)$$

c)

$$\dot{x} = \sin(x)$$

Ex 14.3: Stability of the SIS system [Lvl. 2]

Compute the non-trivial steady state of the SIS model

$$\dot{I} = p(N - I)I - rI$$

and determine its stability.

Ex 14.4: Two abstract two-dimensional system [Lvl. 2]

Compute the steady states of the following systems and analyze their stability:

a) System 1

$$\begin{aligned}\dot{x} &= y^5 - x^2 \\ \dot{y} &= 1 - x\end{aligned}$$

b) System 2

$$\dot{x} = x - y + 2 \tag{54}$$

$$\dot{y} = x - 3y + 8 \tag{55}$$

Ex 14.5: Complex system step-by-step [Lvl. 3]

Let's go step-by-step through the analysis of the following system:

$$\begin{aligned}\dot{x} &= a + bxy \\ \dot{y} &= \frac{1}{2} - x\end{aligned}$$

- Compute the steady state of the system. Then find the Jacobian in the steady state, and compute the eigenvalues.
- Determine for which values of a and b the eigenvalues are complex.
- How does stability depend on a and b if the eigenvalues are complex?
- How does stability depend on a and b if the eigenvalues are real?
- Draw a diagram where a and b are the axis. Use lines to indicate where stability changes or eigenvalues become complex. Color in the area where the steady state is stable.

Ex 14.6: Allee effect [Lvl. 3]

The change in the population size x of an ecological population is described by the differential equation

$$\dot{x} = \frac{x^2}{1 + x^2} - \frac{m}{2}x,$$

where m is a mortality rate.

- Compute the three branches of steady states of this system.
- Determine the stability of the steady states. (This requires a bit more mathematical technique than most exercises in this book. Give it a try. If you get stuck in the roots check the solution.)
- Draw a bifurcation diagram: Plot the x^* over m for $0.5 < m < 1.5$. Indicate the stability of the steady states in the diagram. Also color the area between the branches of steady states according to whether the population size x is increasing or decreasing.
- Now assume we start the system at $m = 0.5$ in a stable steady state with $x^* > 0$. What happens if we slowly increase m ? Once the population goes extinct can we bring it back by decreasing m ? (This is the so-called Allee effect, an example of a tipping point in an ecological system.)

Ex 14.7: Paradox of Enrichment [Lvl. 3]

A fundamental building block of theoretical ecology is the study of predator-prey interaction. A simple model of this interaction is provided by the Rosenzweig-MacArthur system ¹:

$$\begin{aligned}\dot{x} &= 5x \left(1 - \frac{x}{K}\right) - \frac{25xy}{2 + 3x} \\ \dot{y} &= \frac{5xy}{2 + 3x} - y\end{aligned}$$

where x is the population size of the prey population, y is a population size of the predator population and K is the so-called carrying capacity, which can be interpreted as an indicator for the amount of nutrients available to the prey. All three of these quantities are measured in some arbitrary units (e.g. tonnes of biomass, not number of individuals).

- Show that there is one steady state, where neither predator nor-prey survives in the system, and another one where only the prey survives. Then find the steady state in which predator and prey coexist in the system.

¹M. L. Rosenzweig and R. H. MacArthur. Graphical representation and stability conditions of predator-prey interactions. *The American Naturalist*, 97(895):209-223, 1963.

- b) Show that for $0 < K < 1$, the state where the prey is present in the system but the predator is extinct, is the only relevant stable steady state.
- c) If we add nutrients to increase K above 1 then the predator can enter the system. Show that the steady state in which both species coexist, becomes unstable if we raise K beyond a critical point K^* . (This is the paradox of enrichment: Too many nutrients destabilize the system.)

Ex 14.8: Stability condition for maps [Lvl. 4]

In the chapter we derived a general condition for the stability of steady states in ODE systems. Now use the same procedure to find a condition for the stability of fixed points in maps.

- a) Find a stability condition for fixed points in 1-dimensional maps. [Follow the procedure, guessing won't work.]
- b) Now, find the stability condition for multi-dimensional maps $\mathbf{x}_{i+1} = f(\mathbf{x}_i)$, where $\mathbf{x}_i$ is a vector of dimension N .
- c) Suppose you wanted to solve $5x = x^2$ by the fixed-point iteration $x_{i+1} = x_i^2/5$. Do you see why this is problematic? Can you fix it?

Ex 14.9: A puzzle [Lvl. 4]

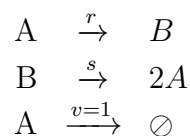
Determine which of the following Jacobians corresponds to a stable steady state (you won't need a computer):

$$\mathbf{J}_1 = \begin{pmatrix} -9 & -1 & 2 & 0 & 1 \\ 0 & -5 & 0 & 1 & 0 \\ 1 & 1 & -4 & 0 & 0 \\ 4 & 1 & 0 & -3 & 0 \\ 3 & 1 & 1 & 1 & -2 \end{pmatrix}, \quad \mathbf{J}_2 = \begin{pmatrix} -4 & 0 & 0 & 1 & 0 \\ 5 & 1 & 0 & 2 & 0 \\ 1 & -2 & -2 & 4 & 4 \\ 6 & 0 & 0 & -1 & 0 \\ -1 & -2 & 0 & 1 & -1 \end{pmatrix}$$

$$\mathbf{J}_3 = \begin{pmatrix} -1 & 0 & 3 & 0 & 0 \\ 0 & 1 & 0 & 0 & 3 \\ -4 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 \\ 0 & 4 & 2 & 1 & -2 \end{pmatrix}, \quad \mathbf{J}_4 = \begin{pmatrix} -2 & 1 & 1 & -4 & 3 \\ 0 & -9 & 1 & 1 & 0 \\ 0 & 0 & -9 & 1 & 0 \\ 0 & 0 & 0 & -9 & 0 \\ 3 & -2 & 2 & 1 & -2 \end{pmatrix}$$

Ex 14.10: Glycolysis [Lvl. 4]

Glycolysis is a process by which your cells turn sugar into energy. The essence of glycolysis can be captured in the following conceptual model:



Here, A describes the amount of available energy. Energy is needed to start a process (first reaction), which then produces more energy (second reaction), which is used to sustain the organism (third reaction).

- a) Let a be the amount of A and b the amount of B in the system. Then, write the ODE system that captures the dynamics of these variables, and show that there's no steady state in which $a^* > 0$ except in a special case.
- b) The absence of a robust steady state where we have a constant level of stored energy is a problem. Let's assume that $r = a^p$, where $p \neq 0$ is a new parameter. Show that there is now a steady state with $a^* > 0$.

- c) Show that the steady state is unstable for $p > 0$. (This leads up to a nice biological insight, check the solution).

Ex 14.11: A simple controller [Lvl. 4]

Suppose we want to stabilize the following system:

$$\begin{aligned}\dot{x} &= f(x, y) = x - 2y \\ \dot{y} &= g(x, y) = 2x + y\end{aligned}$$

- a) Show that the system has a single unstable steady state.
b) We now modify the system by adding a feedback for y . In the simplest case this feedback acts instantaneously and directly such that the equations become

$$\begin{aligned}\dot{x} &= f(x, y) \\ \dot{y} &= g(x, y) + h(y),\end{aligned}$$

where g and y are as above but h is the feedback. Find a function $h(y)$ such that $h(0) = 0$ (to save energy) and the steady state at $(0, 0)$ is stable.

Ex 14.12: The mechanical hydra [Lvl. 5]

This exercise proposes a hypothetical mechanical device. The first subquestions should be easy to solve. The final one has a partial solution. We consider the ODE system

$$\begin{aligned}\dot{x} &= 2x - 4y \\ \dot{y} &= 3x - 5y.\end{aligned}$$

- a) Determine the stability of the steady state at $x = y = 0$.
b) Now imagine that we changed this system by clamping some part in place. As a result y becomes fixed at $y = 0$ and cannot change anymore, which removes it as a variable. Write the differential equation that describes the system after y has been fixed.
c) Determine the stability of the steady state in the resulting system. Is the result surprising?
d) This system exhibits the so-called *hydra effect*: If left alone, it settles into a stable steady state, but if we try to hold a certain part in place in this state, then the rest of the system becomes unstable. Can you design a mechanical gadget, possibly corresponding to a different set of equations, that exhibits a hydra effect?